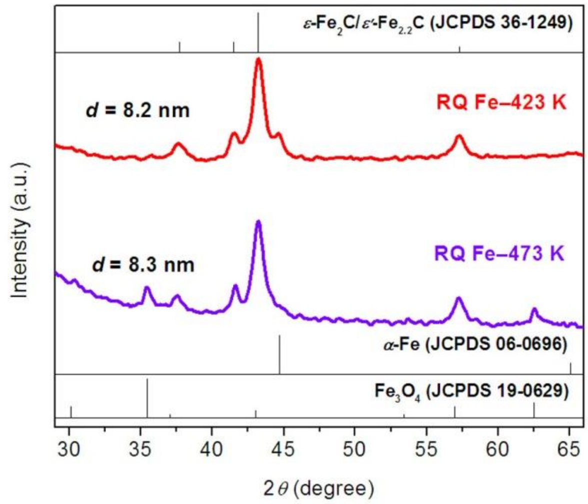
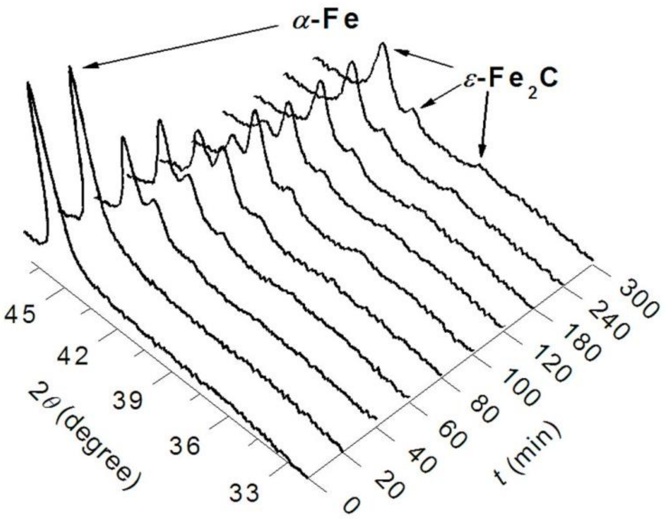
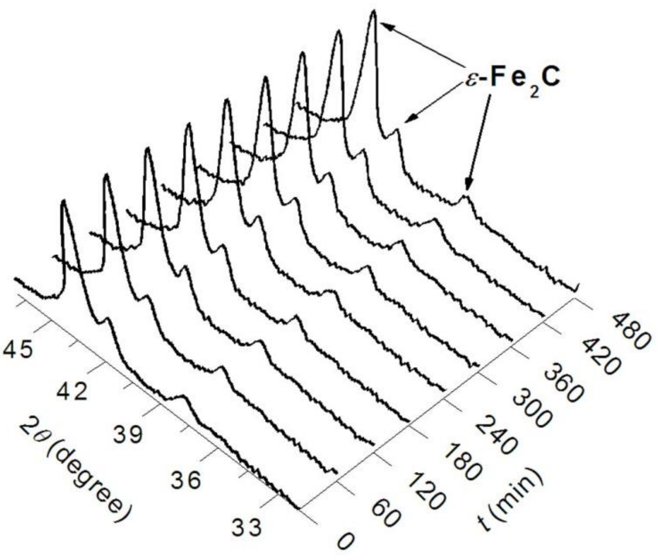


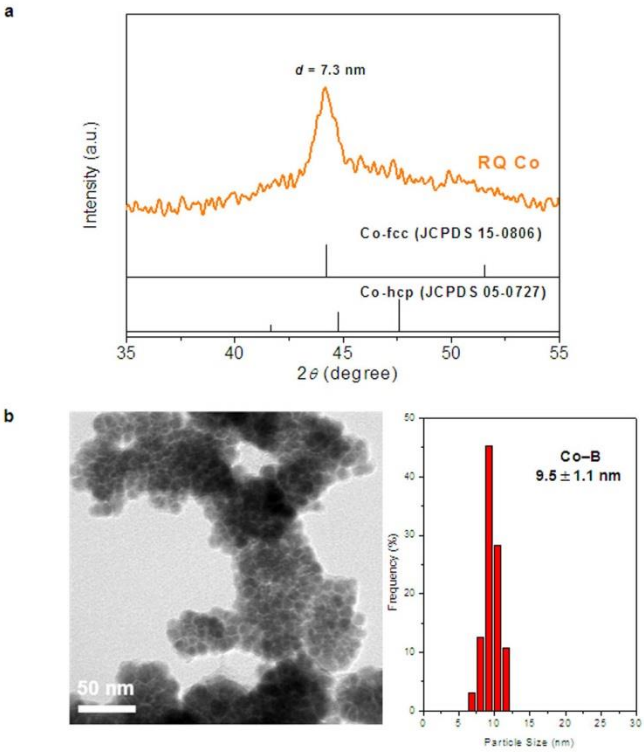
Supplementary Figure 1 | Synchrotron radiation XRD patterns of the RQ Fe after LTFTS at 473 K and 423 K. The predominant phase in the two catalysts is indexable to the O-carbide (JCPDS 36-1249). The crystallite sizes of the O-carbide in the RQ Fe after LTFTS at 473 K and 423 K are estimated to be 8.3 nm and 8.2 nm, respectively, based on the Scherrer equation and the broadening of the primary (101) diffraction peak at 2θ of 43.0°. The RQ Fe after LTFTS at 473 K also contains a small amount of magnetite (Fe_3O_4, JCPDS 19-0629). Instead, the RQ Fe after LTFTS at 423 K contains a small amount of α-Fe (JCPDS 06-0696).



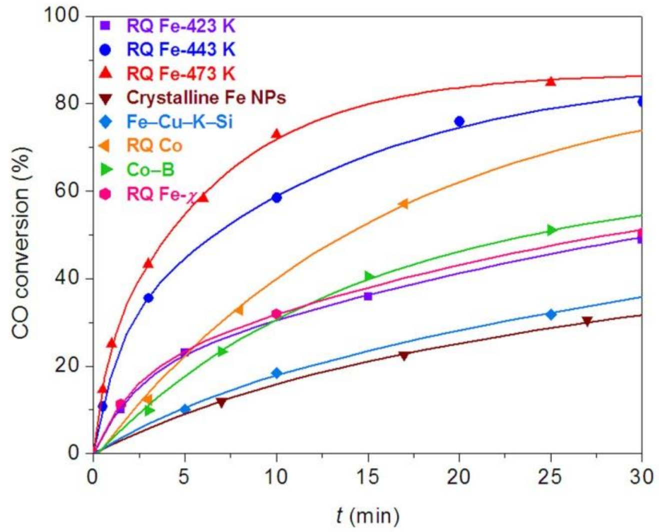
Supplementary Figure 2 | Time-dependent *in situ* XRD patterns at 443 K in syngas for the as-leached RQ Fe. The temperature and gas composition ($H_2/CO/N_2$ = 64/32/4) for *in situ* XRD characterization are the same as those in LTFTS at 443 K. The flow rate and the pressure of the syngas are 20 mL min^{-1} and 5 bar, respectively. According to the figure, with the prolonging of the time, the *α*-Fe phase (JCPDS 06-0696) is consumed gradually, and transformed completely to O-carbide (JCPDS 36-1249) at 180 min. The longer carbidation time than that for on site carbidation in LTFTS at the same temperature may be attributed to the lower pressure adopted during *in situ* XRD characterization. Thereafter, the O-carbide remains as the sole phase with the crystallite size maintaining at 8.3 nm.



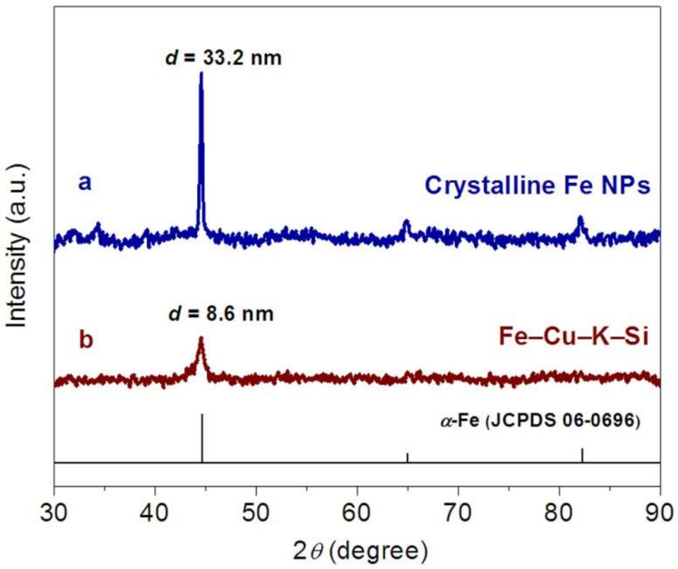
Supplementary Figure 3 | Time-dependent *in situ* XRD patterns at 443 K in syngas for the on site carbided RQ Fe. The RQ Fe is firstly carbided *ex situ* in LTFTS at 443 K in the batchwise reactor for 20 min under the conditions of 30 bar at RT, $H_2/CO/N_2$ of 64/32/4, and PEG200 as the medium. Then, the carbided catalyst is transferred to the *in situ* XRD cell for structural characterization at 443 K in flowing syngas ($H_2/CO/N_2$ = 64/32/4, 20 mL min^{-1}) at 5 bar. According to the figure, the O-carbide (JCPDS 36-1249) remains as the sole phase throughout the measurement, with the peak shape and intensity being unchanged and the crystallite size maintaining at 8.2 nm.



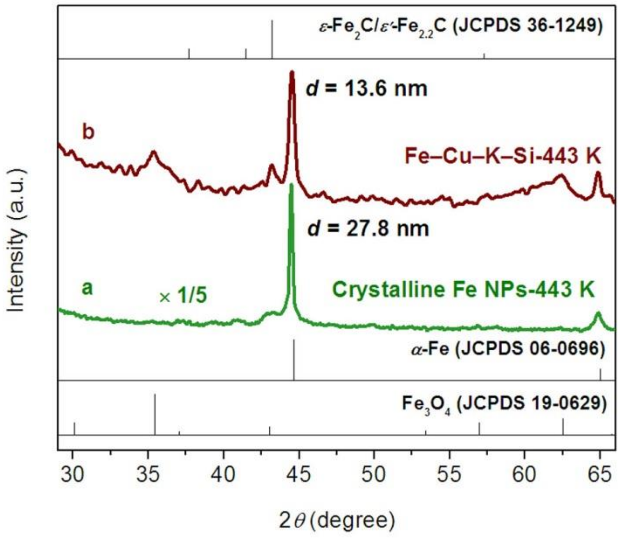
Supplementary Figure 4 | Particle size determination on as-prepared cobalt-based catalysts. a, Powder XRD pattern of the RQ Co catalyst that is indexable to face centred cubic (*fcc*) Co (JCPDS 15-0806); The crystallite size of the *fcc* Co in the RQ Co is estimated to be 7.3 nm. **b**, Transmission electron microscopy (TEM) image and particle size distribution of the amorphous Co–B catalyst; the average particle size of the Co–B is 9.5 nm.



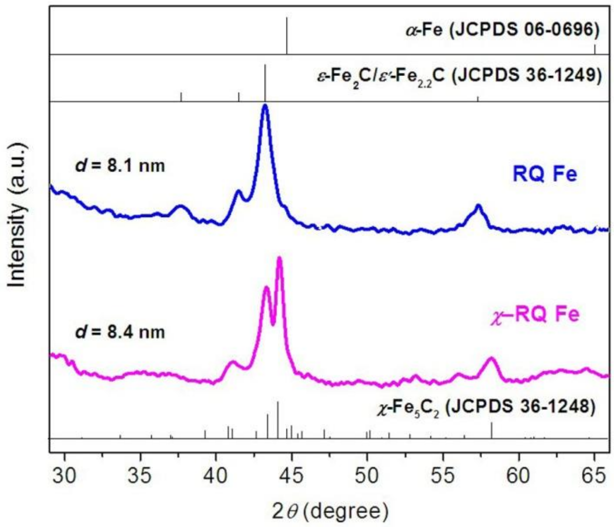
Supplementary Figure 5 | Plots of CO conversion versus reaction time for calculating the initial activities (r_0) of the iron- and cobalt-based catalysts. The r_0 is defined as the amount of CO converted on per mole of iron/cobalt per hour at the beginning of the reaction, which is determined from the initial reaction rate by extrapolating the slope of the CO conversion–time curves to zero reaction time.



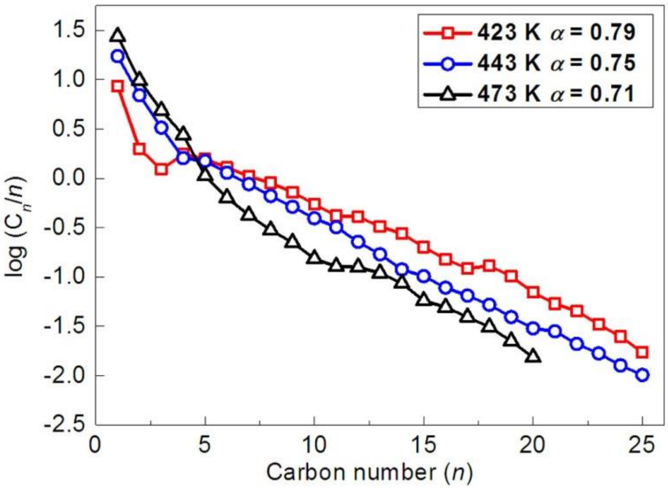
Supplementary Figure 6 | Powder XRD patterns of the comparative crystalline Fe NPs catalyst and the Fe–Cu–K–Si precipitated catalyst. Both catalysts can be well indexed to the α-Fe phase (JCPDS 06-0696). The crystallite sizes are calculated to be 33.2 and 8.6 nm for catalysts (a) and (b), respectively, based on the Scherrer equation and the broadening of the primary (110) diffraction peak.



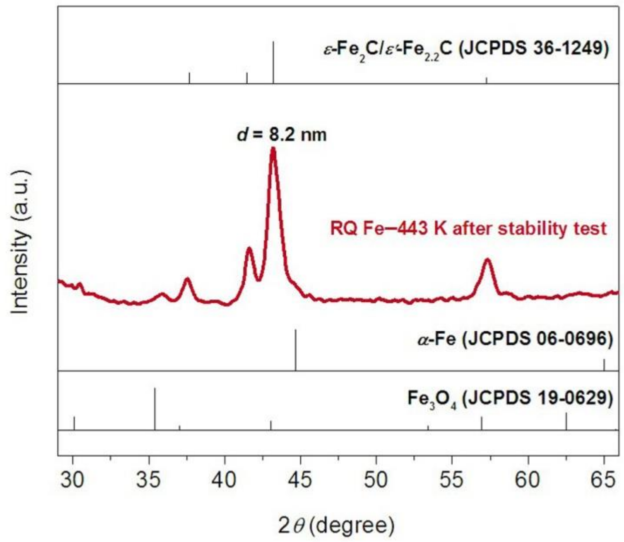
Supplementary Figure 7 | Synchrotron radiation XRD pattern of the crystalline Fe NPs catalyst and the Fe–Cu–K–Si precipitated catalyst after LTFTS at 443 K. For the crystalline Fe NPs catalyst, the majority of α-Fe (JCPDS 06-0696) remains intact after LTFTS, along with a barely visible feature of O-carbide at 2θ of 43.0^{o} (JCPDS 36-1249). The crystallite size of α-Fe decreases to 27.8 nm after LTFTS, signifying the occurrence of surface carbidation on the α-Fe crystallites. The crystallite size of O-carbide is not calculated because of its poorly resolved signal. The Fe–Cu–K–Si precipitated catalyst after LTFTS consists of α-Fe (JCPDS 06-0696), magnetite (JCPDS 19-0629), and O-carbide (JCPDS 36-1249), with the features of the latter two phases being less intense than those of the former. The crystallite size of α-Fe increases to 13.6 nm after LTFTS, and the crystallite size of O-carbide is estimated to be 13.9 nm. It is evident that the crystalline Fe NPs catalyst and the Fe–Cu–K–Si catalyst are more reluctant to carbidation than the RQ Fe catalyst.



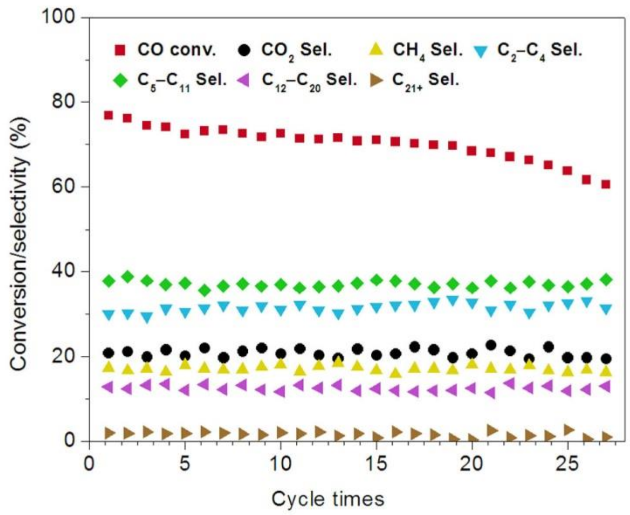
Supplementary Figure 8 | Synchrotron radiation XRD pattern of the comparative RQ Fe-χ catalyst. The RQ Fe-*χ* catalyst can be readily indexed to the *χ*-Fe_5C_2 phase (JCPDS 36-1248). The synchrotron XRD pattern of the RQ Fe catalyst after LTFTS at 443 K is also presented for comparison. Their crystallite sizes and textural properties are compared in Supplementary Table 4.



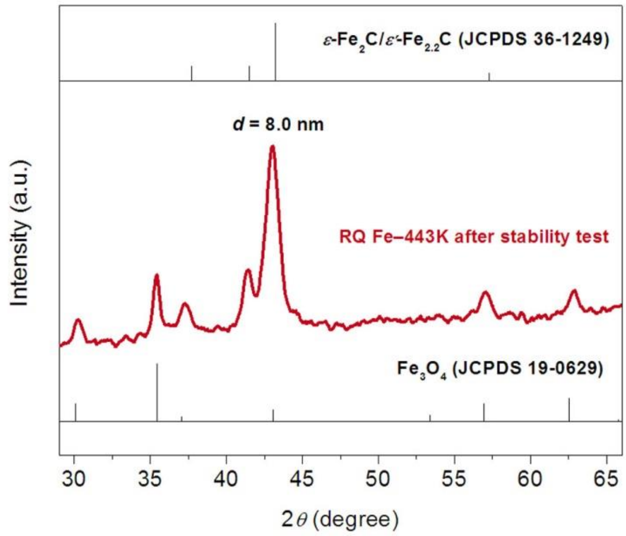
Supplementary Figure 9 | The Anderson–Schulz–Flory (ASF) distributions of the hydrocarbons on the RQ Fe catalyst. Reaction conditions: T = 423–473 K, p = 30 bar at RT, $H_2/CO/N_2$ = 64/32/4, 4.48 mmol Fe, 20 mL PEG200, and stirring rate of 800 rpm. The lower FTS temperature leads to longer hydrocarbons. So the chain growth probability, α, increases steadily from 0.71 at 473 K to 0.79 at 423 K on the RQ Fe catalyst.



Supplementary Figure 10 | Synchrotron radiation XRD pattern of the RQ Fe catalyst after stability test. After five successive runs, ε-Fe_2C remains as the dominant phase in the RQ Fe catalyst. The crystallite size of ε-Fe_2C is 8.2 nm, which increases insignificantly as compared to that of the RQ Fe (8.1 nm) after on site carbidation in LTFTS at 443 K, showing that the ε-Fe_2C phase is highly stable in LTFTS. The emergence of a trace amount of magnetite is possibly due to air exposure during repeated catalyst recovery and/or transfer for stability test and structural characterization.



Supplementary Figure 11 | The CO conversion and product distribution over the RQ Fe catalyst in twenty-seven successive LTFTS cycles. Reaction conditions: T = 443 K, p = 30 bar at RT, $H_2/CO/N_2$ = 64/32/4, 4.48 mmol Fe, 20 mL PEG200, and stirring rate of 800 rpm; after each catalytic run, the catalyst is simply separated from the liquid phase by a magnet, followed by washing with ethanol and PEG200 twice. The recovered catalyst is reused without reactivation. The stability test is intentionally stopped when the CO conversion is decreased to ~60%. The hydrocarbon selectivities are normalized with the exception of CO_2. The hydrocarbon selectivities are normalized with the exception of CO_2. The product distributions are almost unchanged, signifying that the nature of the RQ Fe catalyst does not change during repetitive recycling and stability test.



Supplementary Figure 12 | XRD pattern of the RQ Fe catalyst after twenty-seven successive LTFTS cycles. After the stability test, ε-Fe_2C (JCPDS 36-1249) remains as the dominant phase. The crystallite size of ε-Fe_2C is 8.0 nm, which decreases insignificantly as compared to the size of 8.1 nm of the RQ Fe after on site carbidation in LTFTS at 443 K. The size difference is possibly due to surface oxidation during times of operation of catalyst recovery, as implied by the presence of comparatively distinct features of magnetite (JCPDS 19-0629).

Supplementary Table 1 | Summary of the crystal structures used as input for the calculation of theoretical scattering phases and amplitudes.

	Iron Carbide			
	ε-Fe_2C[a]	**ε'-$Fe_{2.2}C$[b]**	**χ-Fe_5C_2[c]**	**θ-Fe_3C[c]**
ICSD	-	-	76829	38308
Space Group	P6$_3$/*mmc* (194)	P6$_3$/*mmc* (194)	C2/*c* (15)	P*nma* (62)
a	2.749	2.790	11.563	5.092
b	2.749	2.790	4.573	6.741
c	4.340	4.358	5.058	4.527
α	90	90	90	90
β	90	90	97.73	90
γ	120	120	90	90

Atom	**Fe**	**C**	**Fe**	**C**	**Fe (A)**	**Fe (B)**	**Fe (C)**	**C**	**Fe (A)**	**Fe (B)**	**C**
x	0.333	0	0.333	0	0.096	0.215	0	0.115	0.183	0.039	0.876
y	0.667	0	0.667	0	0.088	0.584	0.573	0.304	0.069	0.250	0.250
z	0.250	0	0.250	0	0.418	0.306	0.250	0.084	0.334	0.842	0.443

[a]The structural data of ε-Fe_2C are calculated based on de Smit, E. *et al. J. Am. Chem. Soc.* **132**, 14928–14941 (2010). [b]The structural data of ε'-$Fe_{2.2}C$ are cited from du Plessis, H. E. http://hdl.handle.net/10210/421. [c]The structural data of χ-Fe_5C_2 and θ-Fe_3C are cited from Inorganic Crystal Structure Database (ICSD).

Supplementary Table 2 | Fe K-edge EXAFS fit results of the first Fe–C and Fe–Fe shells and the second Fe–Fe shell of the RQ Fe catalyst after LTFTS at 443 K*.

Pair	***N***	***R* (Å)**	**$\Delta\sigma^2$ (10^{-3} Å^{-2})**	**ΔE_0 (eV)**
Fe–C	1.6	1.91	6.60	1.94
Fe–Fe	4.6	2.73	13.5	10.7
Fe–Fe	5.5	3.90	15.1	-6.10

*N, coordination number; R, distance between absorber and backscatterer atoms; $\Delta\sigma^2$, Debye–Waller factor; and ΔE_0, inner potential correction. Errors: N, ±10%; R, ±0.02 Å.

Supplementary Table 3 | Catalytic activities of the literature iron and ruthenium catalysts and the RQ Fe catalyst in LTFTS.

Entry	Catalyst	T (K)	d[a] (nm)	CO conv. (%)	Activity (mol_{CO} mol_M h^{-1})[b]
1[c]	5% Ru/SiO_2	473	–	3.7–9.8	0.41–1.22
2[d]	Ru nanocluster	423	2.0	75.0	6.9
3[e]	RQ Fe	443	8.1	76	6.7
4[f]	Amorphous Fe-W	473	47.0	84.2	0.83
5[e]	RQ Fe	473	8.3	85	6.0
6[g]	Amorphous Fe NPs	423	8.0	33.0	1.5
7[e]	RQ Fe	423	8.2	36	4.2

[a]Particle size, is determined by TEM for Ru nanocluster, amorphous Fe-W, and amorphous Fe NPs or by XRD for RQ Fe. [b]Activity = (mol CO converted)/(mol M × h) at the CO conversion indicated in the table. M = Fe or Ru. [c]Calculated from Ref. 1. Other reaction conditions: 15 bar, space velocity (H_2 + CO) = 120 mL (NTP) min^{-1} g_{Ru}^{-1}, H_2/CO = 2, and $p(H_2O)$ = 0.17–4.54 bar. The particle size of 5% Ru/SiO_2 was not mentioned in the literature. [d]Cited from Ref. 2. Other reaction conditions: 30 bar, H_2/CO = 2, 0.279 mmol Ru, PVP/Ru = 20/1, 20 mL H_2O, and stirring rate of 800 rpm. [e]This work. Other reaction conditions: 30 bar, $H_2/CO/N_2$ = 64/32/4, 4.48 mmol Fe, 20 mL PEG200, and stirring rate of 800 rpm. [f]Cited from Ref. 3. Other reaction conditions: 30 bar, H_2/CO = 2, 6 mmol Fe, 40 g PEG200, and stirring rate of 800 rpm.

[g]Cited from Ref. 4. Other reaction conditions: 30 bar, H_2/CO = 2, 2 mmol Fe, 40 mL PEG200, and stirring rate of 800 rpm.

Supplementary Table 4 | The crystallite sizes and textural properties of the ε-Fe_2C-dominant RQ Fe catalyst and the χ-Fe_5C_2-dominant RQ Fe-χ catalyst.

Catalyst	$d_{cryst.}$[a]	S_{BET}[b] ($m^2\ g^{-1}$)	V_{pore}[b] ($cm^3\ g^{-1}$)	d_{pore}[b] (nm)
RQ Fe	8.1	41	0.19	19.3
RQ Fe-χ	8.3	45	0.19	17.9

[a]Crystallite size calculated by the Scherrer equation. [b]Determined by N_2 physisorption at 77 K on Micromeritics TriStar3000 apparatus.

Supplementary Methods

Reagents: $FeCl_3 \cdot 6H_2O$ (99.0%), $Fe(NO_3)_3 \cdot 9H_2O$ (98.5%), $Co(CH_3COO)_2 \cdot 4H_2O$ (99.5%), $Cu(NO_3)_2 \cdot 3H_2O$ (99.0%), Na_2CO_3 (99.8%), $K_2SiO_3 \cdot xH_2O$ (80 wt% K_2SiO_3, CP grade), KBH_4 (95.0%), $N_2H_4 \cdot H_2O$ (85.0%), PEG200 (CP grade), cyclohexane (99.5%), and decahydronaphthalene (99.5%) were purchased from Sinopharm. KOH (85.0%) was purchased from Shanghai Lingfeng. HNO_3 (65–68%) was purchased from Chinasun Specialty. Ethanol (99.7%) was purchased from Shanghai Zhenxing. The RQ $Co_{40}Al_{60}$ alloy (Co/Al, *w*/*w*) was prepared in a way similar to the RQ $Fe_{50}Al_{50}$ alloy.

Calculation of the activity. For calculating the initial activity, r_0, the experimental CO conversion–time curve is fitted by Origin 8 program to derive the relationship between the CO conversion and reaction time:

$$X_{CO} = f(t) \tag{1}$$

where X_{CO} denotes the conversion of CO determined gas chromatographically at time t in unit h.

Then, Eq. 1 is differentiated with respect to t to derive the function of the slope k:

$$k = f'(t) \tag{2}$$

By substituting 0 for t, one obtains k_0, i.e., $\frac{dX_{CO}}{dt}|_{t=0}$, that represents the slope of the CO conversion–time curve at zero reaction time. Hence, the initial activity, r_0 (unit in $mol_{CO}\ mol_M\ h^{-1}$), is calculated as:

$$r_0 = \frac{k_0 \times n_{CO}}{n_M} \tag{3}$$

where n_{CO} is the molar number of CO and n_M (M = Fe, Co, or Ru) is the molar number of active metal charged in the batch reactor, respectively.

For calculating the activity at a certain X_{CO} (denoted as Activity), the corresponding reaction time t is read out from the experimental CO conversion–time curve or figured out from Eq. 1. Hence, the Activity (unit in mol_{CO} mol_{M} h^{-1}) is calculated as:

$$\text{Activity} = \frac{X_{CO} \times n_{CO}}{n_{M} \times t} \tag{4}$$

where n_{CO} and n_{M} are the same as those defined below Eq. 3.

Supplementary References

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